

Shraddha Shinde

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SUMMARY

Senior RF & Electromagnetics Engineer with extensive experience (6+ years) in simulation-driven design and validation of RF components and systems for medical, defense, and satellite applications. Deep expertise in CST Studio Suite, HFSS, ADS, and MATLAB, covering antenna, filter, and RF subsystem development from concept through compliance and production. Recognized for bridging theoretical electromagnetics with practical hardware implementation and cross-functional collaboration.

Skills & Core Competencies

- **RF & Electromagnetic Design:** MRI, RADAR, Satellite systems
- **Simulation Tools:** CST Studio Suite, HFSS, ADS, FD3D, MATLAB, Antenna Magus, LTspice
- **PCB Design:** Altium Designer, Gerber files, BOM, Cadence Allegro
- **Testing & Measurement:** Vector Network Analyzer, Spectrum Analyzer, Oscilloscope, Signal Generator, RF Power Meter
- **Compliance & Validation:** IEC 60601-1/-2, DFMEA, DFM, CAPA, Safety Assessments
- **PLM & Tools:** Windchill PLM, Jira
- **Programming:** Python, Embedded C, MATLAB
- **Documentation:** Technical documentation, white paper publications, audit reports
- **Collaboration & Leadership:** Customer/stakeholder engagement, intern mentoring, vendor coordination

Professional Experience: Industry (RF / Electromagnetics)

1. Philips, Pune, INDIA

Senior Electrical Engineer, MRI Research and Development Department (NPI Projects). *August 2023 – March 2025*

- **Designed compact MRI RF coils using HFSS and CST, performing impedance matching and S-parameter validation.**
- Developed PCBAs in Altium Designer, managing layout, component selection, and vendor coordination.
- Led design validation (simulation, DFMEA, DFM), ensuring IEC 60601-1/-2 compliance.
- Implemented CAPA and root cause analysis for quality and reliability improvements.
- Acted as Bucket Leader for customer feedback, enhancing complaint resolution.
- Authored technical documentation, design records, white papers, and audit-ready change request storyboards.
- Managed product lifecycle, verification, and compliance documentation via Windchill PLM.

2. Dassault Systemes, Pune, INDIA

Electromagnetic Specialist, Research and Development.

June 2021 - August 2023

- Proficient in EM/CAD simulation using CST Studio Suite and Filter Designer 3D (FD3D).
- Designed RF front-end modules (filters and antennas) for high-frequency applications.
- Developed Diplexer and Triplexer circuits with application notes/documentation records for end users.
- Implemented cavity, planar, and circuit filters using FD3D and full-wave simulations.
- Modelled array patch antennas and microstrip filters for satellite communication.
- Reported bugs via JIRA, collaborating with the development team.
- Assisted 150+ customers in troubleshooting high-frequency electromagnetic issues in CST software.
- Managed team events, overseeing planning and execution.

3. Exicom Technology (INDIA) LLP, Mumbai, INDIA

Research and Development Engineer (Radio Frequency (RF)) (Contract position)

July 2020 - June 2021

- Prepared documentation for PCB assembly (Altium software), schematics, and fabrication, Gerber files and BOMs.
- Calibrated lab equipment, including Vector Network Analyzer.
- Trained interns on SDR (Software Defined Radio) integration and testing, improving project outcomes.
- Created Bill of Materials (BOM) for SDR integration and testing.
- Developed a work management system, increasing team productivity by over 60%.

4. SAMEER (Society for Applied Microwave Electronics Engineering and Research, IIT Bombay, Government Sector) (Contract position)

1. Research and development (M.Tech Associate)

July 2019 - July 2020

- **Project:** - RF-based Microstrip Hairpin Bandpass filter technique for Radio Detection and Ranging (RADAR) applications at the X band.
- Designed, optimized, and simulated RF microstrip bandpass filters using CST, ADS, and Genesys.
- Fabricated filters on substrate and Tested on a Vector Network Analyzer.
- Coordinated with vendors for fabrication and maintained documentation records.
- Calculated X-band filter parameters using MATLAB and Python.

2. Research and development Associate

November 2016 - November 2017

- **Project:** -Portable Tabletop Moisture Measurement uses a Microwave guide/Microstrip Patch Antenna at 5.81GHz.
- Designed and simulated a microstrip patch antenna using HFSS for EM wave transmission and reception.
- Tested on a Vector Network Analyzer with documentation. Trained 5 colleagues on the installation of halogen and resistance application devices.

Professional Development

March 2025 – July 2025

- Completed certifications in Project Management, RF Design, PCB Layout, IoT, and Wireless Technologies.
- Enhanced skills in Python, MATLAB, and Electromagnetics through online learning and lectures.

Selected Teaching Experience (Industry-Aligned)

Fr. Conceicao Rodrigues College of Engineering(FRCRCE), INDIA, Mumbai

July 2025 – Present

Assistant Professor (Adhoc, Computer Science and Electronics Department)

- Applied MATLAB and simulation-based workflows to explain signal processing, electromagnetics, and electronic system behavior.
- Guided final-year engineering projects with focus on system modeling, algorithm implementation, and technical documentation aligned with industry practices.
- Integrated simulation tools and hands-on labs to strengthen students understanding of hardware and software co-design.
- Conducted hands-on laboratory experiments using Function Generator, Digital Multimeter, and Digital Storage Oscilloscope to demonstrate practical circuit analysis and signal measurement techniques.

Academic & Freelance Experience

1. SaiYash Tutorial, Mumbai, INDIA

Freelancer: Lecturer (On-site, Part-Time)

May 2016 - June 2023

- Tutored 40+ undergraduates in Electronics, achieving a 25% year-on-year increase in average results.
- Managed conflicts between students and instructors, initiating family interactions for additional support when needed.

2. Subject Matter Expert (Electronics & Electrical) Chegg India (Remote, Part-time)

March 2020 - September 2025

- Solved 200+ advanced questions in electronics, electromagnetics and electrical circuits.
- Maintained a 90%+ quality rating, providing clear, step-by-step solutions for global undergraduate students.

EDUCATION

Veermata Jijabai Technological Institute (VJTI) Aggregate CGPA - 7.62/10

July 2018- August 2020

Master of Technology in Electronics and Telecommunication Engineering

- Researched microstrip hairpin bandpass filter techniques, achieving a 60% size reduction and improved roll-off for narrow-band RADAR applications.

St. Francis Institute of Technology, Mumbai University Aggregate CGPA - 6.63/10

June 2013- June 2016

Bachelor of Engineering in Electronics and Telecommunication

- Completed a team project on "IRIS Recognition using match score fusion" using MATLAB during Bachelor of Engineering.
- Implemented a microstrip stub filter for L-band in ADS software for medical applications.

Additional qualification: Mumbai University (Correspondence-IDOL)

June 2016 - June 2018

Masters of English Literature

- Learned about Indian : American, and Anglo-Irish literature in English, Linguistic and Stylistic Analysis of texts.

Vidyalankar Polytechnic Technology, MSBTE Aggregate (%) - 80.80/100

June 2010 - June 2013

Diploma in Electronics and Telecommunication

- Conducted a project on "Fire Extinguisher" using ultraviolet and flame sensors, involving circuit design, simulation with MATLAB, and PCB development in eagle software.

PUBLICATIONS

- Shraddha Shinde, Arun Heddallikar and D.P.Rathod, "Design and Simulation of Microstrip Bandpass Filter Techniques at X band for RADAR Applications", Springer Journal paper published, ICTCS, Smart Trends for computing and communication, December 17-18. 2021. https://link.springer.com/chapter/10.1007/978-981-16-9967-2_6.
- Shraddha Shinde, Arun Heddallikar and D.P.Rathod, "A Comparative Study of 5th Order Microstrip Hairpin and Parallel Coupled Bandpass Filter with Various Microstrip Filter Techniques", ICTELCON 2020, February 28. 2020. <https://www.tcetmumbai.in/Multicon/2020/ICTELCON%202020.pdf>
- Shraddha Shinde, Arun Heddallikar, and D.P. Rathod, "Survey of Microstrip based Bandpass Filter Techniques with Development of Filters for Various Applications", IEEE Hyderabad Section, Electromagnetics and Microwave Engineering Pedagogy, Research Trends and Applications (EMPRA 2020), January 2. 2020.
- E health care service for rural population, National Level Technical Paper Presentation, State Level Technical Paper Presentation, March – December. 2012.

LANGUAGES

English (Full Professional proficiency), **Marathi** (Full Professional proficiency), **Hindi** (Full Professional proficiency), **German** (Elementary Proficiency), **French** (Intermediate Proficiency-A1).

CERTIFICATION AND LICENSE

- **AI for All: AI Aware and AI Appreciate** - Intel and Government of India (Digital India).
- **Career Essentials in Project Management** - Microsoft & LinkedIn.
- **Python Programming Essentials** - Rice University.
- **Digital Thread: Components** - University at Buffalo.
- **Digital Manufacturing & Design** - University at Buffalo.
- **Smartphone Emerging Technologies** - Yonsei University.
- **Wireless Communication Emerging Technologies** - Yonsei University.
- **Internet Emerging Technologies** - Yonsei University.
- **Internet of Things: Communication Technologies** - University of California San Diego.
- **Altium Education PCB Basic Design Course** - Training in PCB design using Altium Tools.
- **Business Analysis and Process Management** - Coursera.
- **Critical Thinking and Problem Solving** - LinkedIn.
- **Matlab Onramp/Simulink** - MATLAB.

EXTRACURRICULAR ACTIVITIES

- Attended a workshop on Mathematics in Engineering: A student-driven Pedagogical Endeavour, IIT B, TEQIP, and Initiative of the MHRD, Govt. of India, October. 2018.
- Our (Mother and Me) efforts and achievements grabbed the attention of "The Logical Indians Magazine", which got Spotlited on the 21st of April. 2017. <https://thelogicalindian.com/my-story/my-story-shraddha-shinde/>.
- MARATHON Winner in Gulf Petroleum (7Km) Night Race (2017, 2018, 2019).
- Enthusia Marathon 2025 (5km) National level completed.

VOLUNTEERING

AKSHARA (Government Authorized NGO) towards Gender Justice (Counselor and Active Member)

Justice for Women, Children, and Senior Citizens

June 2010 - Present

- Held a volunteer mentor position in events, was an interviewer on the interview panel for membership of girls (45+girls) for the scholarship and gained excellent leadership and management qualities by mentoring 10 individuals.

Teach for India (Teaching Assistant)

February 2018 - April 2018

Teaching Underprivileged School Children

- Assisted in teaching and educating underprivileged students (70+ students) aged 12-14 years old, so that they can get an equal chance to face the world in the education sector.

Rangoonwala Foundation Trust (Student Volunteer)

November 2018 – January 2023

Social cause, health care and welfare

- Worked on spreading awareness of TB and Citizen Rights in Hospitals among people (300+) by conducting presentations among the community in public as well as in institutions and organizations.